

UNZENDAKE, Japan

WMO: 478180

Lat: 32.7378N

Lon: 130.2625E

Elev: 680

StdP: 93.42

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.6	-3.2	-9.9	1.8	-0.2	-8.4	2.0	-0.5	12.0	5.4	10.3	2.5	4.2	300	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.8	28.5	23.2	27.5	22.9	26.5	22.5	24.2	26.6	23.7	25.8	23.2	25.0	2.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.1	25.2	23.1	19.5	24.5	22.7	18.9	24.1	77.4	26.9	75.1	26.0	73.1	25.2	26.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.8	10.7	9.1	-6.6	30.4	1.4	1.0	-7.7	31.1	-8.5	31.7	-9.3	32.3	-10.4	33.0		
			-7.5	25.6	1.4	0.7	-8.5	26.1	-9.4	26.5	-10.2	26.8	-11.2	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.2	2.4	3.7	6.7	11.6	16.0	19.3	22.9	23.8	20.7	15.7	10.1	4.7
	DBStd	7.87	3.22	3.65	3.42	3.14	2.20	1.82	1.67	1.54	2.37	2.84	3.51	3.51
	HDD10.0	757	236	179	111	20	0	0	0	0	0	1	42	168
	HDD18.3	2319	493	410	359	203	76	11	0	0	7	89	247	423
	CDD10.0	1920	1	2	10	67	187	279	400	427	322	178	45	3
	CDD18.3	441	0	0	0	0	4	40	142	168	78	8	0	0
	CDH23.3	1508	0	0	0	0	13	47	491	746	203	9	0	0
	CDH26.7	187	0	0	0	0	1	1	65	108	12	0	0	0
Wind	WSAvg	3.6	3.6	3.7	3.9	3.8	3.5	3.4	3.7	3.4	3.5	3.5	3.4	3.8
Precipitation	PrecAvg	2555	79	114	183	208	233	510	437	247	231	116	108	89
	PrecMax	4225	179	251	417	361	470	1008	997	686	628	287	241	166
	PrecMin	1396	13	52	93	89	61	123	59	53	51	10	20	2
	PrecStd	541	38	48	68	78	105	223	237	151	148	77	58	43
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	15.1	17.2	21.6	25.0	26.0	29.6	30.1	28.0	24.6	19.5	14.9
		MCWB	12.4	12.7	12.0	14.2	15.8	19.4	23.3	23.6	22.1	19.2	15.7	12.2
	2%	DB	11.0	13.3	15.3	19.4	23.0	24.7	28.3	28.8	26.7	22.7	18.1	13.3
		MCWB	9.5	11.7	11.5	13.5	15.6	19.0	23.2	23.3	21.7	18.5	15.3	11.1
	5%	DB	9.4	11.7	13.8	18.0	21.7	23.5	27.3	27.9	25.6	21.4	16.8	12.0
		MCWB	7.7	10.1	11.1	13.8	15.5	19.3	23.0	23.1	21.3	17.8	14.7	10.0
	10%	DB	7.7	9.8	12.4	16.7	20.5	22.6	26.1	26.9	24.4	20.3	15.6	10.5
		MCWB	5.7	7.9	10.2	12.9	15.5	19.8	22.6	22.9	20.8	17.1	13.4	8.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.6	14.3	15.0	18.1	20.3	22.7	25.1	25.2	23.8	21.6	17.8	14.0
		MCDB	13.0	14.4	15.2	18.6	20.9	23.1	27.5	27.7	25.7	22.7	18.3	14.1
	2%	WB	10.2	12.2	13.3	16.7	19.2	22.2	24.2	24.5	23.0	20.4	16.5	12.0
		MCDB	10.7	12.7	14.0	17.6	20.0	22.6	26.5	27.0	24.6	21.7	17.2	12.7
	5%	WB	8.0	10.4	11.9	15.3	18.2	21.7	23.6	23.9	22.4	19.2	15.1	10.4
		MCDB	8.9	11.4	12.9	16.3	19.3	22.2	25.6	26.3	24.1	20.5	16.3	11.5
	10%	WB	6.0	8.4	10.6	14.1	17.2	21.1	23.1	23.5	21.8	18.0	13.8	8.8
		MCDB	7.3	9.5	12.0	15.7	18.9	21.8	24.7	25.6	23.6	19.5	15.3	10.2
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.9	7.5	7.9	8.0	5.8	5.1	5.8	6.4	7.2	6.9	6.4
		MCDBR	8.0	8.5	9.0	9.3	10.2	8.0	7.6	7.6	7.6	8.2	7.1	7.8
		MCWBR	7.1	7.6	6.9	6.0	5.9	4.5	4.0	3.8	4.0	4.9	5.1	6.4
	5% WB	MCDBR	7.3	7.6	7.4	6.3	6.5	3.8	6.2	6.5	5.9	5.6	5.9	6.8
		MCWBR	7.4	7.7	7.1	5.7	5.4	3.1	3.7	3.5	3.5	4.3	5.2	6.6
Clear-Sky Solar Irradiance	taub	0.409	0.447	0.503	0.521	0.530	0.539	0.530	0.507	0.477	0.432	0.404	0.400	
	taud	2.078	2.011	1.869	1.876	1.915	1.977	2.052	2.114	2.147	2.182	2.180	2.117	
	Ebn at Noon	777	789	774	782	778	768	772	785	791	797	780	762	
	Edh at Noon	132	156	193	200	194	182	168	156	145	129	118	120	
All-Sky Solar Radiation	RadAvg	2.27	2.98	4.04	4.82	5.14	4.26	5.00	5.23	4.31	3.62	2.66	2.11	
	RadStd	0.23	0.28	0.26	0.42	0.58	0.66	0.68	0.48	0.46	0.37	0.26	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page